

NEURAL DISTINGUISHER PROJECT DOCUMENTATION

Deep Learning & Classical ML Framework for Cryptographic Algorithm Identification from Ciphertext

1. Executive Summary & Problem Formulation

Modern block ciphers are mathematically engineered to produce high-entropy pseudo-random noise that satisfies Shannon's principle of confusion and diffusion. Under traditional cryptanalysis, ciphertext without knowledge of the cryptographic key should be computationally indistinguishable from an ideal random permutation.

The **Neural Distinguisher** project investigates whether deep representation learning and classical machine learning can extract non-randomness from 49 NIST SP 800-22 statistical randomness metrics to identify the underlying encryption algorithm across two problem domains:

- **Binary Classification (AES vs. 3DES):** A focused discriminator assessing structural block-size and diffusion differences between AES-128 and Triple-DES.
- **5-Cipher Multiclass Classification:** A broader evaluation across 5 distinct ciphers (AES-128, 3DES, CAST-128, RC2, Blowfish) evaluating cross-cipher statistical separability against random chance (20.0%).

2. End-to-End Pipeline Architecture (5 Core Stages)

The system follows an automated pipeline divided into five flow stages:

- **Stage 1 - Simulation & Data Generation:** Generates 25,000 synthetic ciphertext samples across AES, 3DES, CAST, RC2, and Blowfish in ECB/CBC modes with PKCS#7 padding across five scale tiers: 1 KB, 8 KB, 64 KB, 256 KB, and 512 KB.
- **Stage 2 - NIST Statistical Feature Extraction:** Computes 49 statistical randomness p-values per sample based on NIST SP 800-22 tests (Monobit, Block Frequency, Runs, Longest Run of Ones, Discrete Fourier Transform, Maurer's Universal, Approximate Entropy, and Cumulative Sums).
- **Stage 3 - Multi-Architecture Model Training:** Trains 7 distinct model families: 1D Convolutional Neural Network (CipherCNN), Multi-Layer Perceptron (MLP 64x32), Random Forest (100 trees), Support Vector Machine (RBF kernel), K-Nearest Neighbors (k=5), Logistic Regression, and Gaussian Naive Bayes.
- **Stage 4 - Checkpoint Verification & Packaging:** Implements 10-split stratified shuffle validation, loss checkpointing, feature standardization, and artifact export.
- **Stage 5 - Tester & Empirical Benchmark Evaluation:** Conducts empirical benchmarking, automated visualization, and live multi-architecture inference.

3. Model Architecture Specifications

The system benchmarks seven diverse learning architectures to evaluate feature extraction capabilities:

- **1D CNN (Proposed Model):** A deep convolutional network with two Conv1D layers (16 and 32 filters, kernel size 3, padding 1), MaxPool1D pooling, ReLU activations, and a 64-unit fully-connected dense layer with Dropout.
- **MLP (Deep Baseline):** Two fully-connected hidden layers (64 and 32 neurons) with ReLU activation and Adam optimization.
- **Random Forest:** An ensemble of 100 decorrelated decision trees capturing non-linear feature interactions.
- **Support Vector Machine (SVM):** Non-linear hyperplane separation using a Radial Basis Function (RBF) kernel with C=1.0.
- **K-Nearest Neighbors (KNN):** Instance-based metric classification using Euclidean distance across standardized features (k=5).
- **Logistic Regression (LR):** Linear discriminative baseline using L2 ridge regularization and L-BFGS optimization.
- **Gaussian Naive Bayes (GNB):** Probabilistic baseline assuming independent normal feature distributions.

4. Empirical Benchmark Performance

All models were evaluated across 10 repeated stratified splits. Below are the verified empirical benchmarks for both classification tasks:

[A] Binary Classification Benchmark: AES vs. 3DES (Random Chance = 50.0%)

The 1D CNN achieves up to **92.0% accuracy** at 512 KB, outperforming classical machine learning baselines by up to 38.0%:

File Size	1D CNN	MLP	Random Forest	SVM	KNN	Log. Reg.	GNB
1 KB	82.0%	72.5%	52.5%	50.0%	55.0%	40.0%	44.0%
8 KB	84.5%	72.5%	57.5%	52.5%	52.5%	50.0%	52.0%
64 KB	89.5%	77.5%	65.0%	62.5%	60.0%	60.0%	60.0%
256 KB	89.5%	77.5%	62.5%	57.5%	57.5%	54.0%	62.0%
512 KB	92.0%	82.5%	60.0%	60.0%	60.0%	54.0%	60.0%

[B] 5-Cipher Multiclass Classification Benchmark (Random Chance = 20.0%)

Evaluated across AES-128, 3DES, CAST-128, RC2, and Blowfish on the 25,000-sample empirical dataset:

File Size	1D CNN	MLP	Random Forest	SVM	KNN	Log. Reg.	GNB
1 KB	19.4%	19.4%	21.8%	18.5%	18.9%	18.6%	18.3%
8 KB	20.1%	19.6%	18.6%	19.3%	20.7%	18.0%	18.1%
64 KB	22.0%	20.0%	19.0%	20.2%	20.7%	21.0%	20.6%
256 KB	20.2%	21.4%	19.5%	19.9%	21.2%	19.8%	20.1%
512 KB	20.2%	19.9%	19.9%	18.9%	22.5%	21.1%	18.8%

Key Finding: While 1D CNN successfully distinguishes AES from 3DES binary pairs due to structural differences, multi-cipher statistical discrimination hovers between **18.0% and 22.5%** around the **20.0% random-chance level**. This confirms the cryptographic principle that secure block ciphers produce statistically indistinguishable pseudo-random distributions across 5 algorithms.

5. Interactive System & Production Features

The production inference interface (predict.py) and REST API provide the following robust features:

- **Multi-Format Input Ingestion:** Supports raw hexadecimal strings, bulk CSV batch uploads, and real-time synthetic generation.
- **Dynamic Input Snapping:** Inputs are aligned to trained benchmark sizes without zero-padding, preserving ciphertext entropy.
- **Weighted Probability Consensus:** Ensemble prediction weighting the 1D CNN at 2.0x and classical baselines at 1.0x to prevent majority vote dilution.
- **Out-of-Distribution Rejection:** Inputs that fail entropy checks or match non-target distributions are rejected rather than misclassified.
- **Automated Visual Reporting:** Automatically creates publication-grade visual charts for both binary (benchmark_comparison.png) and multiclass (multiclass_benchmark_comparison.png) performance.